Lesson 9

COMPUTER SIMULATION MODELING & NEUROIMAGING

The emergence of various types of computer science technology has brought a lot of changes to how people do things everyday. It has changed the way of living, priorities, and perspectives of people. For instance, people turn to the Internet and other computing technologies hoping to satisfy their curiosity about human behavior and mental processes. They read, listen, and browse the Internet for specific topics such as abnormal, social, cognitive, and developmental psychology. With the Internet, it is possible to access almost any information.

However, the Internet has also brought challenges and changes to research. The way researches are conducted two decades ago is entirely different from the present. Today, there are many procedures by which researchers obtain information about a phenomenon using various types of computer science technology and other electronic devices. This chapter will briefly discuss computer systems by which researchers can acquire knowledge in a scientific approach. This chapter defines computer simulation modeling and neuroimaging. It also presents the uses and methods of computer simulation modeling. The chapter concludes with the different techniques of neuroimaging.

Use of Computer Automation in Research

As early as the 1970s, researchers were making use of computers in psychological experiments to perform tasks such as delivering a standardized and controlled presentation of stimuli and making accurate recordings of responses. Since then, the number of experiments using the Internet and various other types of computing technology has grown considerably. In the 1990s, the move to conduct human psychological experimentation on the Internet was made possible by the development of a hypertext transfer protocol or http. This allowed an Internet browser, such as Netscape Navigator or Internet Explorer, to get a document it located on a server. This document, or Web page, is coded in a language known as hypertext markup language, or HTML, and this language permits the display of text, graphics, or other information on a Web page. With the ability to display such a combination of words, pictures, and sounds on Web pages, the Web grew at an astonishing rate. At this time, most human experimental research in psychology is aided by computer automation (Christensen, Johnson, & Turner, 2015).

Computer Simulation Modeling

Researchers use models in an attempt to gain understanding and insights about some aspect of the real world. A model is a surrogate system for another system. There can be many reasons for using a model. For instance, models can enable researchers to study how a prospective system will work before the real system has even been built. Models have a common purpose which is to mimic or describe the behavior of the system being modeled. A computer simulation is a model which happens to be a computer program (Sanchez, 2007). Simulation refers to any procedure that is meant to imitate a real-life system. Simulations are especially useful in examining situations that are too complex and too costly to investigate in the real world. The computer is often used for such studies because itis able to efficiently model systems and process data.

The *Encyclopedia Britannica* defines computer simulation as the use of a computer to represent the dynamic responses of one system by the behavior of another system modeled after it. A simulation uses a mathematical description or model of a real system in the form of a computer program. Computer simulations are used to study the dynamic behavior of objects or systems in response to conditions that cannot be easily or safely applied in real life. Simulations are especially useful in enabling observers to measure and predict how the functioning of an entire system may be affected by altering individual components within that system.

Computer simulations are run on a computer and uses step-by-step methods to explore the approximate behavior of a mathematical model. It is a comprehensive method for studying systems. In this broader sense of the term, it refers to an entire process that includes choosing a model; finding a way of implementing that model in a form that can be run on a computer; calculating the output of the algorithm; and visualizing and studying the resultant data. The method includes this entire process-used to make inferences about the target system that one tries to model-as well as the procedures used to sanction those inferences (Winsberg, 2013).

In the behavioral sciences, simulation studies have been used to provide evidence of the performance of statistical methods under a variety of data conditions. The goal is to encourage methodological researchers to treat these studies as statistical sampling experiments subject to established principles of experimental design and data analysis. Experimental design should play a crucial role in simulation studies because of its ability to produce effects of interest, guide analyses of study outcomes, and enhance generalizability of findings (Harwell, Kohli, & Peralta, 2017).

The Use of Computer Simulation Modeling

There are three general purposes of computer simulations. Simulations can be used for heuristic purposes (Winsberg,2013). Computer simulations can be used for learning, discovery, or problem-solving using a heuristic computer program. It allows people to solve problems and make judgments quickly and efficiently. For instance, researchers interested in human cognitive processes have long used computer simulations to try to identify the principles of cognition. The strategy has been to build computational models that embody a set of principles and then examine how well the models capture human performance in cognitive tasks (Plaut, 2000).

Psychologists seek to understand the factors that influence human behaviors under a particular set of circumstances. Computer simulations allow psychologists to derive explanations and predictions about behaviors. Researchers use computer simulated models to predict the future, and infer or explain past events. Finally, simulations can be used to understand systems and their behavior. If there are already data to tell how some system behaves, computer simulation allows researchers to derive explanations about how events could possibly have occurred or about how those events actually did occur (Winsberg, 2013).

Methods of Computer Simulation

Monte Carlo Simulation

In a Monte Carlo simulation, values for uncertain variables are generated by the computer to reproduce information found in the real world. It is named after the city of Monte Carlo, Monaco where the primary attractions are games of chance at gambling casinos. Much like the random behavior in games of chance, a Monte Carlo simulation selects values at random to simulate a variable. In a Monte Carlo simulation, behavioral processes are entirely simulated by the computer.

Virtual Reality Simulation

Virtual reality simulation is a computer-simulated environment in which the user of the virtual reality computer technology experiences telepresence. Telepresence refers to a feeling present in an environment that is generated by a communication medium such as a computer. Virtual reality simulation reproduces the experience of reality as participants navigate through the simulated environment. The simulation is typically multidimensional in which the simulated environment attempts to arouse and full range of and participants' senses such as visual, olfactory, auditory, and tactile.

Microworld Simulation

Microworld simulations are complex, computer-generated situations used in controlled experiments that are designed to study decision-making. In a microworld simulation, the participants are instructed to make a decision as they navigate through a computer-generated decision-making situation.

Lesson 9

Neuroimaging

Neuroimaging or brain imaging is the use of various techniques to either directly or indirectly image the structure and functions of the nervous system. A brain imaging method could be defined as any experimental technique that allows human brain structures or functions to be studied. Researchers developed imaging techniques that use the computer's ability to generate images of the parts of the brain from sources of radiation (Rathus, 2017).

Techniques of Neuroimaging

Neuroimaging techniques are largely classified into two categories: techniques that measure the electric activity of cell groups in the brain, such as electroencephalography (EEG) and magnetoencephalography (MEG), and techniques that measure the change in blood flow associated with brain activity, such as functional magnetic resonance imaging (fMRI), near-infrared spectroscopy (NIRS), and position emission tomography (PET).

EEG is a technique used to measure the natural electric activity in the brain using electrodes attached to the scalp. MEG is used to accurately measure the change of magnetic field produced by electric activity in the brain. Structural MMRI provides information to describe the shape, size, and integrity of gray and white matter structures in the brain. In addition to the structural images of the brain, the functional images of the brain can be also visualized by using the fMRI. The fMRI technique can be used not only to identify brain regions active when participants perform a task that requires a particular neuronal process, but also to examine functional connectivity across multiple brain regions while participants are not performing a particular task. NIRS is a technique used to measure changes in oxygenated and deoxygenated hemoglobin in specific brain areas in response to various stimuli (Morita, Asada, & Naito, 2016).The PET forms a computer-generated image of the activity of parts of the brain by tracing the amount of glucose used by these parts (Rathus, 2017).

Key Points

- Simulation refers to any procedure that is meant to imitate a real-life system.
- Computer simulation is the use of a computer to represent the dynamic responses of one system by the behavior of another system modeled after it.
- Computer simulations can be used for heuristic purposes, to derive explanations and predictions about behaviors, and to understand systems and their behavior.
- Neuroimaging or brain imaging is the use of various techniques to either directly or indirectly image the structure and functions of the nervous system.
- Neuroimaging techniques are classified into two categories: one is the technique to measure the electric activity of cell groups in the brain and the other is the technique to measure the change in blood flow associated with brain activity.

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